北京大學
PEKING UNIVERSITY

第三章 分子结构

1. 分子轨道理论

（一） 概述

分子的薛定谔方程

$$\hat{H}\Psi = E\Psi, \qquad \hat{H} = T_n + T_e + V_{en} + V_{ee} + V_{nn}$$

$$\hat{H} = -\sum_{A=1}^{N_n} \frac{1}{2M_A}\nabla^2_{\boldsymbol{R}_A} - \sum_{i=1}^{N_e} \frac{1}{2}\nabla^2_{\boldsymbol{r}_i} - \sum_{i=1}^{N_e}\sum_{\mu=1}^{N_n} \frac{Z_\mu}{|\boldsymbol{r}_i - \boldsymbol{R}_\mu|}$$
$$+ \sum_{i=1}^{N_e}\sum_{j>i}^{N_e} \frac{1}{|\boldsymbol{r}_i - \boldsymbol{r}_j|} + \sum_{\mu=1}^{N_n}\sum_{\nu>\mu}^{N_n} \frac{Z_\mu Z_\nu}{|\boldsymbol{R}_\mu - \boldsymbol{R}_\nu|}.$$

波恩—奥本海默 (BO) 近似 (1927)

$$\hat{H}_{\mathrm{BO}} = \sum_{i=1}^{N_e} -\frac{1}{2}\nabla^2_{\boldsymbol{r}_i} - \sum_{i=1}^{N_e}\sum_{A=1}^{N_n} \frac{Z_A}{|\boldsymbol{r}_i - \boldsymbol{R}_A|} + \sum_{i=1}^{N_e}\sum_{j>i}^{N_e} \frac{1}{|\boldsymbol{r}_i - \boldsymbol{r}_j|} + \sum_{A=1}^{N_n}\sum_{B>A}^{N_n} \frac{Z_A Z_B}{|\boldsymbol{R}_A - \boldsymbol{R}_B|}.$$

BO问题

$$\hat{H}_{\mathrm{BO}}(\underline{\underline{\boldsymbol{r}}};\underline{\underline{\boldsymbol{R}}})\Phi_k(\underline{\underline{\boldsymbol{r}}};\underline{\underline{\boldsymbol{R}}}) = \epsilon_k^{\mathrm{BO}}(\underline{\underline{\boldsymbol{R}}})\Phi_k(\underline{\underline{\boldsymbol{r}}};\underline{\underline{\boldsymbol{R}}}).$$

$$\underline{\underline{\boldsymbol{r}}} = (\boldsymbol{r}_1, \boldsymbol{r}_2, \cdots, \boldsymbol{r}_{N_e}), \quad \underline{\underline{\boldsymbol{R}}} = (\boldsymbol{R}_1, \boldsymbol{R}_2, \cdots, \boldsymbol{R}_{N_n}).$$

$\epsilon_k^{\mathrm{BO}}(\underline{\underline{\boldsymbol{R}}})$: adiabatic potential energy surfaces

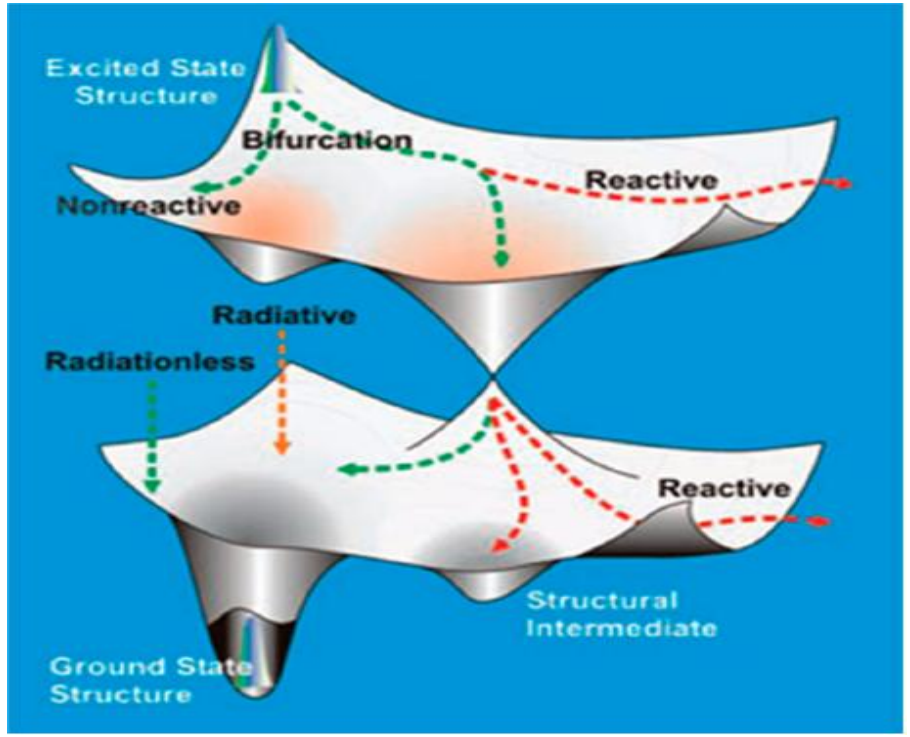
BO势能面对于平动、转动具有不变性

$$\epsilon_k^{\mathrm{BO}}(\underline{\underline{\boldsymbol{R}}}) = \epsilon_k^{\mathrm{BO}}(\underline{\underline{\boldsymbol{R}}}').$$

$$\underline{\underline{\boldsymbol{R}}} = (\boldsymbol{R}_1, \boldsymbol{R}_2, \cdots, \boldsymbol{R}_{N_n})$$

$$\underline{\underline{\boldsymbol{R}}}' = (\boldsymbol{R}_1 + \boldsymbol{S}, \boldsymbol{R}_2 + \boldsymbol{S}, \cdots, \boldsymbol{R}_{N_n} + \boldsymbol{S})$$

因此势能面常用内坐标表示（去掉平转自由度）



R. Srinivasan, J. S. Feenstra, S. T. Park, S. Xu, A. H. Zewail, Science 2005, 307, 558–563.

以水分子为例，核心任务：求解**薛定谔**方程（能量、波函数）

$$\hat{H}\Psi = E\Psi$$

$$\hat{H} = \sum_{i=1}^{N_e} -\frac{1}{2}\nabla_{\boldsymbol{r}_i}^2 - \sum_{i=1}^{N_e}\sum_{\mu=1}^{N_n}\frac{Z_\mu}{|\boldsymbol{r}_i - \boldsymbol{R}_\mu|} + \sum_{i=1}^{N_e}\sum_{j>i}^{N_e}\frac{1}{|\boldsymbol{r}_i - \boldsymbol{r}_j|} + \sum_{\mu=1}^{N_n}\sum_{\nu>\mu}^{N_n}\frac{Z_\mu Z_\nu}{|\boldsymbol{R}_\mu - \boldsymbol{R}_\nu|}$$

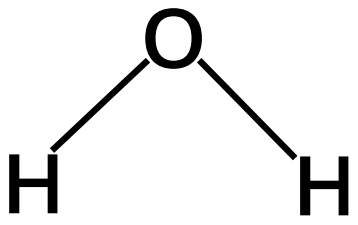
有相互作用项，无法分离变量

常数，可以后加

- 水分子，$N_e = 10$，$N_n = 3$
- 原子核位置是**常量**，电子位置是**变量**

$$\Psi = \Psi(\boldsymbol{r}_1, \boldsymbol{r}_2, \boldsymbol{r}_3, \cdots, \boldsymbol{r}_{10})$$

$$= \Psi(x_1, y_1, z_1, x_2, y_2, z_2, \cdots, x_{10}, y_{10}, z_{10})$$

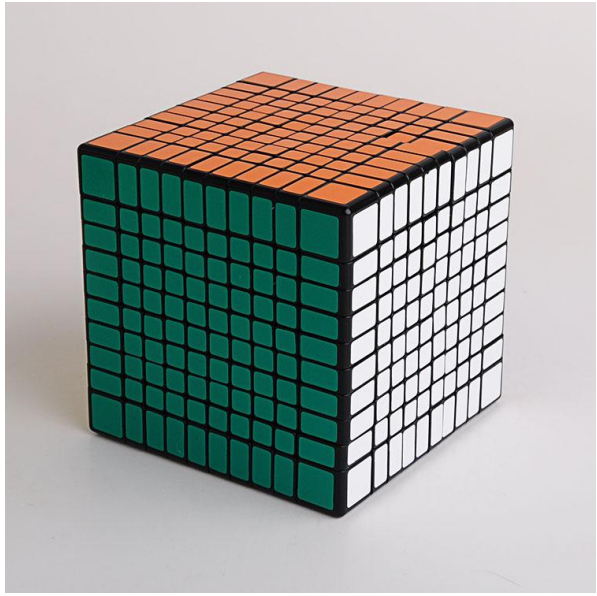


- 波函数是30个变量的函数！ → **维度灾难**

估计一下水分子波函数的信息量

$$\Psi = \Psi(\boldsymbol{r}_1, \boldsymbol{r}_2, \boldsymbol{r}_3, \cdots, \boldsymbol{r}_{10})$$

$$= \Psi(x_1, y_1, z_1, x_2, y_2, z_2, \cdots, x_{10}, y_{10}, z_{10})$$



用离散格点表示，每一个维度用11个点，

$$x_i = \{-1, -0.8, -0.6, \cdots, 1\}$$

11^{30} $\xrightarrow{\text{反对称性}}$ $\dfrac{11^{30}}{10!} = 4.8 \times 10^{24}$ $\xrightarrow{\text{1Byte/数据}}$

$4.8 \times 10^{24}\mathrm{B} = 4.8 \times 10^{15}\mathrm{GB} = 4.8 \times 10^{12}\mathrm{TB}$ $\xrightarrow{\text{5TB/移动硬盘}}$ 10^{12} 硬盘

$\xrightarrow{\text{200g/硬盘}}$ $2 \times 10^{14}\mathrm{g} = 2 \times 10^{8}\mathrm{t}$ **2亿吨！** 2022年上海港年吞吐量为7亿吨

🤦 😂

吾之计算力也有涯，
而计算量也无涯，
以有涯追无涯，
则殆已！

思考题：如能穷尽宇宙中所有物质制作存储设备，最大能存下什么原子的波函数？

1929 狄拉克：大部分物理和**全部化学**的基本规律现在已完全知道了，困难只是在于应用这些规律所得到的方程太复杂，无法求解

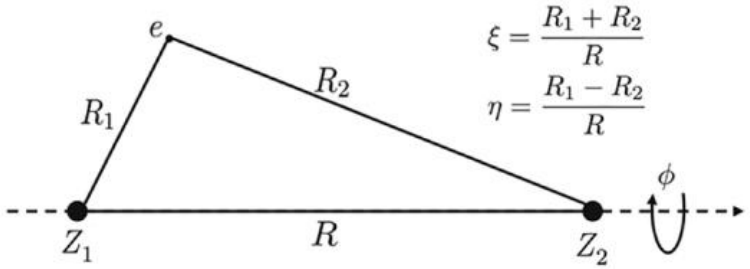
五十年代 鲍林：也许我们可以相信理论物理学家，物质的所有性质都应当用薛定谔方程来算，但实际上，自从薛定谔方程发现的30年来，我们看到化学家感兴趣的物理性质只有很少几个做出了准确而非经验性的量子力学计算。

（二） 氢分子离子

氢分子离子H_2^+

BO近似，原子单位下薛定谔方程

$$-\frac{1}{2}\nabla^2\psi+\left(-\frac{1}{R_1}-\frac{1}{R_2}+\frac{1}{R}\right)\psi=E\psi$$



单电子方程，本征波函数即**分子轨道**

方程具有对称性，可转换到**长椭球坐标系**(spheroidal coordinates)

转换到长椭球坐标系后

$$\nabla^2=-\frac{4}{R^2(\xi^2-\eta^2)}\left[\frac{\partial}{\partial\xi}\left((\xi^2-1)\frac{\partial}{\partial\xi}\right)+\frac{\partial}{\partial\eta}\left((1-\eta^2)\frac{\partial}{\partial\eta}\right)+\left(\frac{1}{\xi^2-1}+\frac{1}{1-\eta^2}\right)\frac{\partial^2}{\partial\phi^2}\right]$$

$$-\frac{1}{R_1}-\frac{1}{R_2}=-\frac{2}{R}\left(\frac{1}{\xi+\eta}+\frac{1}{\xi-\eta}\right)=-\frac{4\xi}{R(\xi^2-\eta^2)}$$

令 $\tilde{E}=E-\frac{1}{R}$

代入原方程，两边乘以 $\frac{1}{4}R^2(\xi^2-\eta^2)$

$$-\frac{1}{2}\left[\frac{\partial}{\partial\xi}\left((\xi^2-1)\frac{\partial}{\partial\xi}\right)+\frac{\partial}{\partial\eta}\left((1-\eta^2)\frac{\partial}{\partial\eta}\right)+\left(\frac{1}{\xi^2-1}+\frac{1}{1-\eta^2}\right)\frac{\partial^2}{\partial\phi^2}\right]\psi-R\xi\psi=\frac{1}{4}R^2(\xi^2-\eta^2)\tilde{E}\psi$$

令 $\psi=N(\xi)M(\eta)\frac{1}{\sqrt{2\pi}}e^{im\phi}$ 可实现变量分离

$$\left[\frac{\partial}{\partial\eta}\left((1-\eta^2)\frac{\partial}{\partial\eta}\right)-\frac{m^2}{1-\eta^2}-\frac{R^2}{2}\tilde{E}\eta^2-A\right]M(\eta)=0,$$

$$\left[\frac{\partial}{\partial\xi}\left((1-\xi^2)\frac{\partial}{\partial\xi}\right)-\frac{m^2}{1-\xi^2}-\frac{R^2}{2}\tilde{E}\xi^2-2R\xi-A\right]N(\xi)=0.$$

椭球坐标 $\longleftrightarrow$ 球坐标

$\xi \geqslant 1 \longleftrightarrow r$

$-1 \leqslant \eta \leqslant 1 \longleftrightarrow \cos\theta$

方程中$\tilde{E}$与A（与角动量以及Runge-Lenz矢量有关）

是待定常数，将两个方程耦合

$\alpha_2 = \sqrt{-\frac{R^2}{2}\tilde{E}}$

类氢原子：e^{-Zr}

调制函数，无穷级数

基态正解

$$\psi = C\xi^{\gamma}\exp[-\alpha_1\eta - \alpha_2\xi + F(\eta,\xi)]$$

一般分子轨道

$$\psi_{KLm}(\eta,\xi,\phi) = C\left[\sqrt{(1-\eta^2)(\xi^2-1)}\right]^{|m|} e^{im\phi} \qquad (m \neq 0 \text{ 相关部分})$$

$$\times\left[\prod_{l=1}^{L}(\eta-a_l)\right]\left[\prod_{k=1}^{K}(\xi-b_k)\right] \qquad (\text{波函数的节面})$$

$$\times\,\xi^{\gamma}e^{-\alpha_1\eta-\alpha_2\xi+F(\eta,\xi)}$$

Y. Li, C. Li, ACS Omega, 7, 22594, 2022.

传统基组展开解法

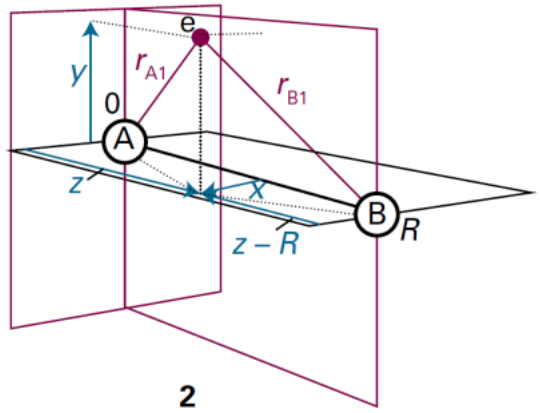
原子轨道线性组合（linear combination of atomic orbitals, LCAO）

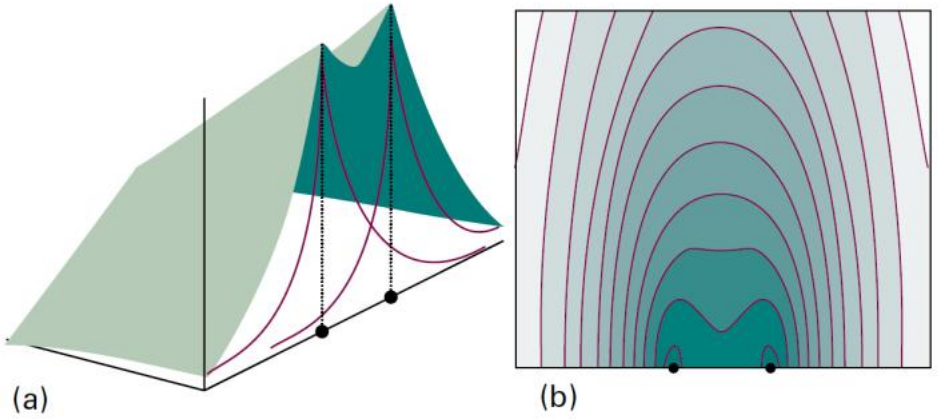
$$\psi_{\pm} = N_{\pm}(\psi_A \pm \psi_B)$$

ψ_A, ψ_B为原子轨道(这里只考虑1s轨道)

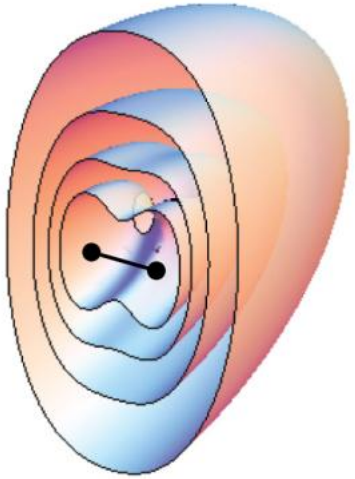
基态叫做**σ轨道：沿键轴方向像s轨道，**

角动量分量为0 (m=0)





基态波函数在xz平面的图像



基态波函数空间等值面

成键轨道

轨道密度 $\psi_+^2 \propto \psi_A^2 + \psi_B^2 + 2\psi_A\psi_B$

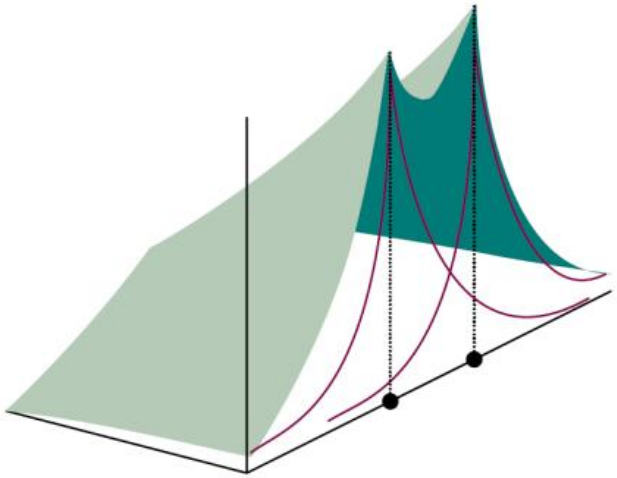
交叠密度 $2\psi_A\psi_B$

交叠密度导致**核间区域干涉相长，电子密度增加**

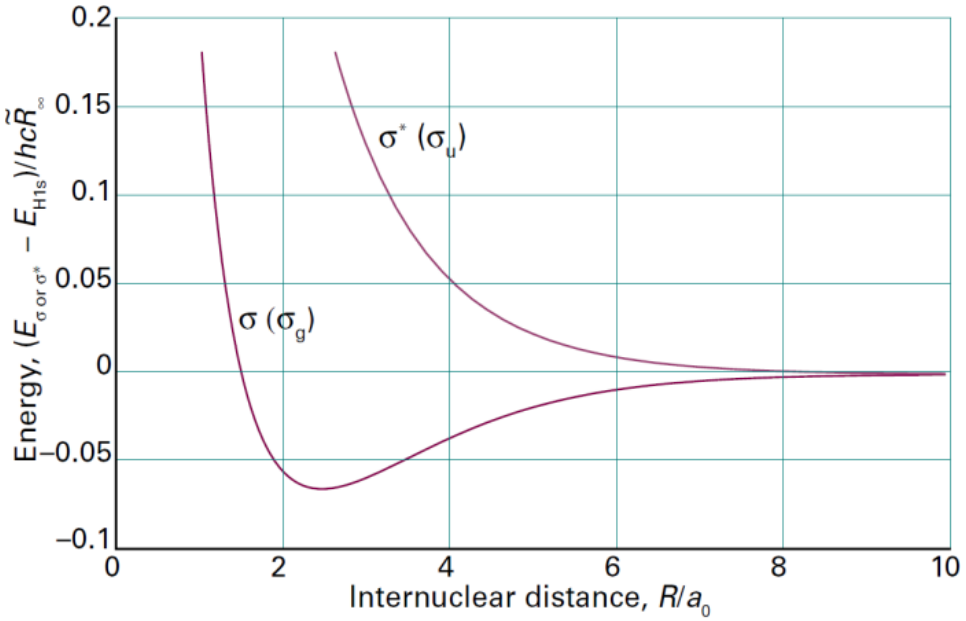
电子密度视角理解成键：**核间密度累积**

能量视角理解成键：**能量降低**
(相对解离情况)

思考：电子密度视角与能量视角有什么关联？能否由一个视角解释另一个视角？



基态电子密度



LCAO推导基态能量

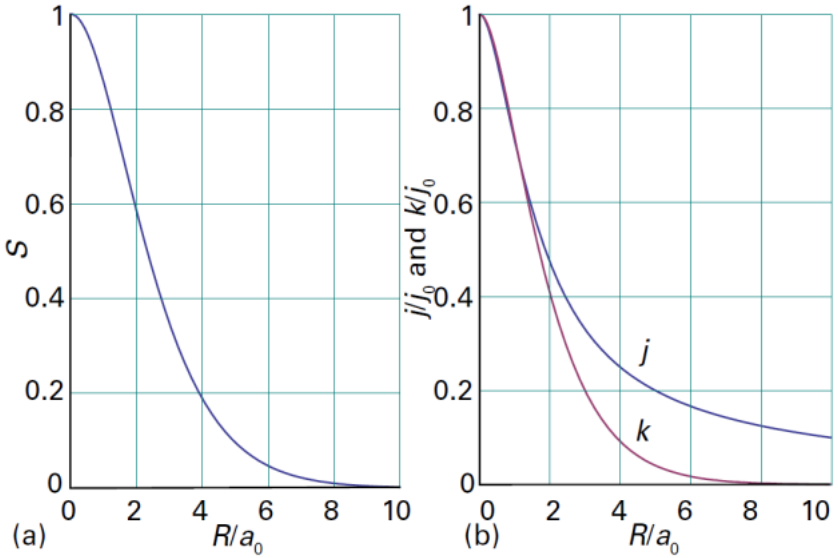
图像定性正确（键长132pm/106pm，D_e 1.76eV/2.6eV）　思考：误差来源

$$E_\sigma = E_{H1s} + \frac{j_0}{R} - \frac{j+k}{1+S}$$

交叠积分　$$S = \int \psi_A \psi_B \, d\tau = \left\{1 + \frac{R}{a_0} + \frac{1}{3}\left(\frac{R}{a_0}\right)^2\right\} e^{-R/a_0}$$

库伦积分　$$j = j_0 \int \frac{\psi_A^2}{r_B} \, d\tau = \frac{j_0}{R}\left\{1 - \left(1 + \frac{R}{a_0}\right) e^{-2R/a_0}\right\}$$

库伦交叠积分　$$k = j_0 \int \frac{\psi_A \psi_B}{r_B} \, d\tau = \frac{j_0}{a_0}\left(1 + \frac{R}{a_0}\right) e^{-R/a_0}$$



S与k指数衰减，j库伦衰减

j_0是能量单位　$$\frac{j_0}{a_0} = \frac{e^2}{4\pi\varepsilon_0 a_0} = \frac{e^2}{4\pi\varepsilon_0} \times \frac{\pi m_e e^2}{\varepsilon_0 h^2} = \frac{m_e e^4}{4\varepsilon_0^2 h^2} = 2hc\tilde{R}_\infty$$

反键轨道σ^*

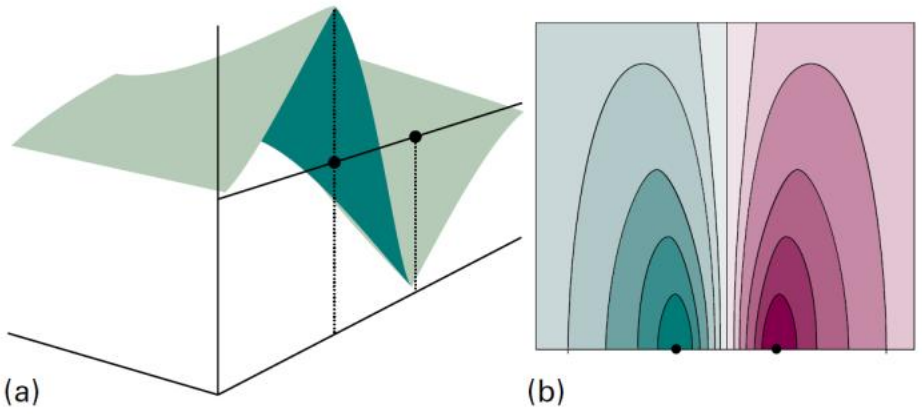
轨道密度 $\psi_-^2 \propto \psi_A^2 + \psi_B^2 - 2\psi_A\psi_B$

交叠密度 $-2\psi_A\psi_B$

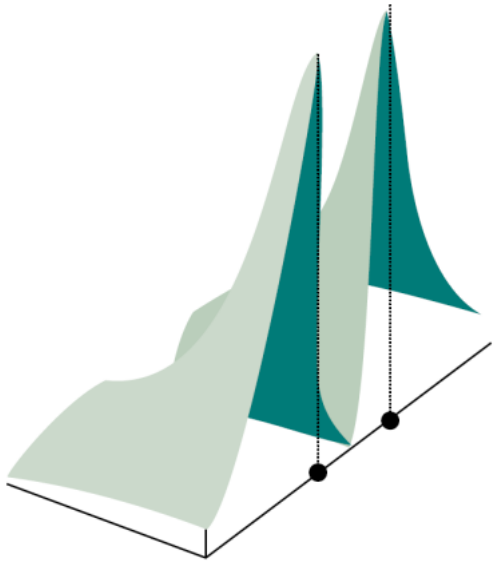
交叠密度导致**核间区域干涉相消，电子密度减小，出现节面**

电子密度视角理解反键：**核间密度减小，核外密度累积**

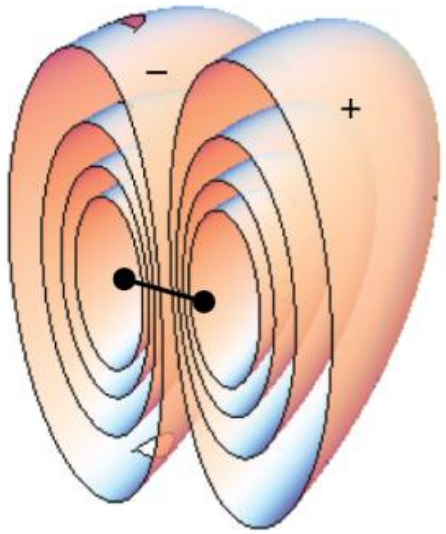
能量视角理解反键：**能量升高(相对解离情况)**



σ^*波函数在xz平面的图像

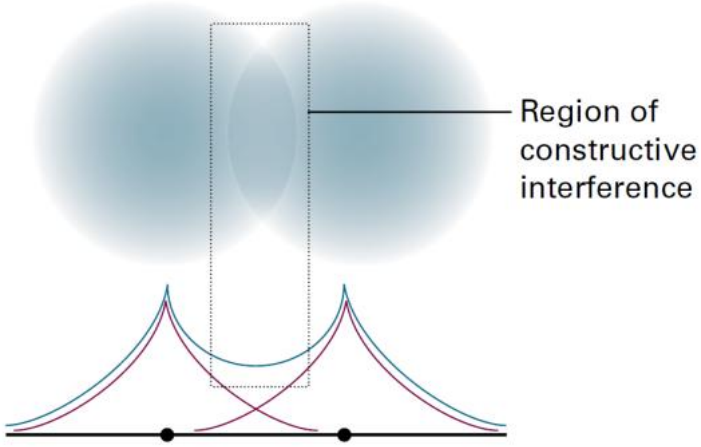


反键轨道密度



反键轨道空间等值面

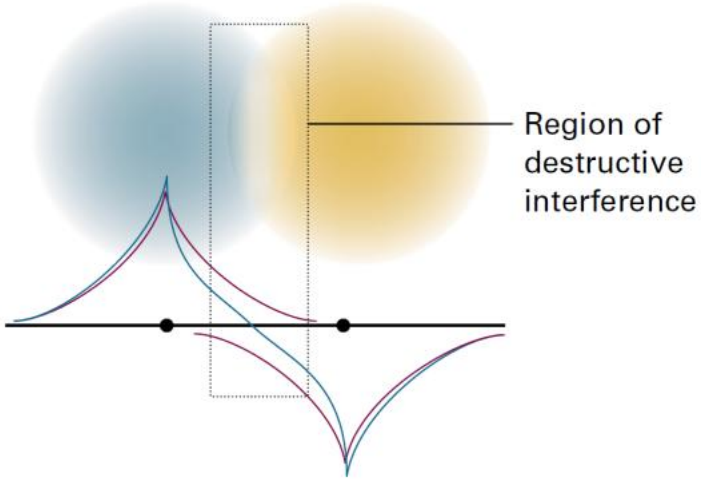
成键轨道σ



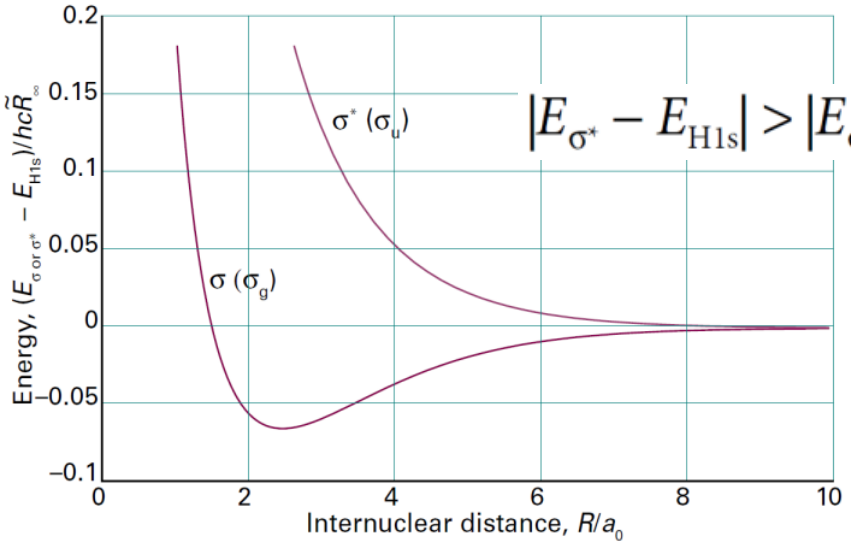
$$E_{\sigma}=E_{\mathrm{H1s}}+\frac{j_0}{R}-\frac{j+k}{1+S}$$

能量曲线一般会
出现极小值

反键轨道σ^*



$$E_{\sigma^*}=E_{\mathrm{H1s}}+\frac{j_0}{R}-\frac{j-k}{1-S}$$



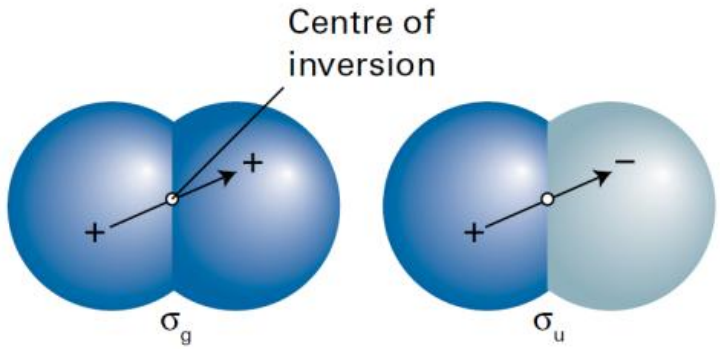
$$|E_{\sigma^*}-E_{\mathrm{H1s}}|>|E_{\sigma}-E_{\mathrm{H1s}}|$$

能量曲线一般
单调下降

反键带来的能量升高比
成键带来的能量降低更多

为什么？

轨道宇称(parity) = 反演对称性(inversion symmetry)

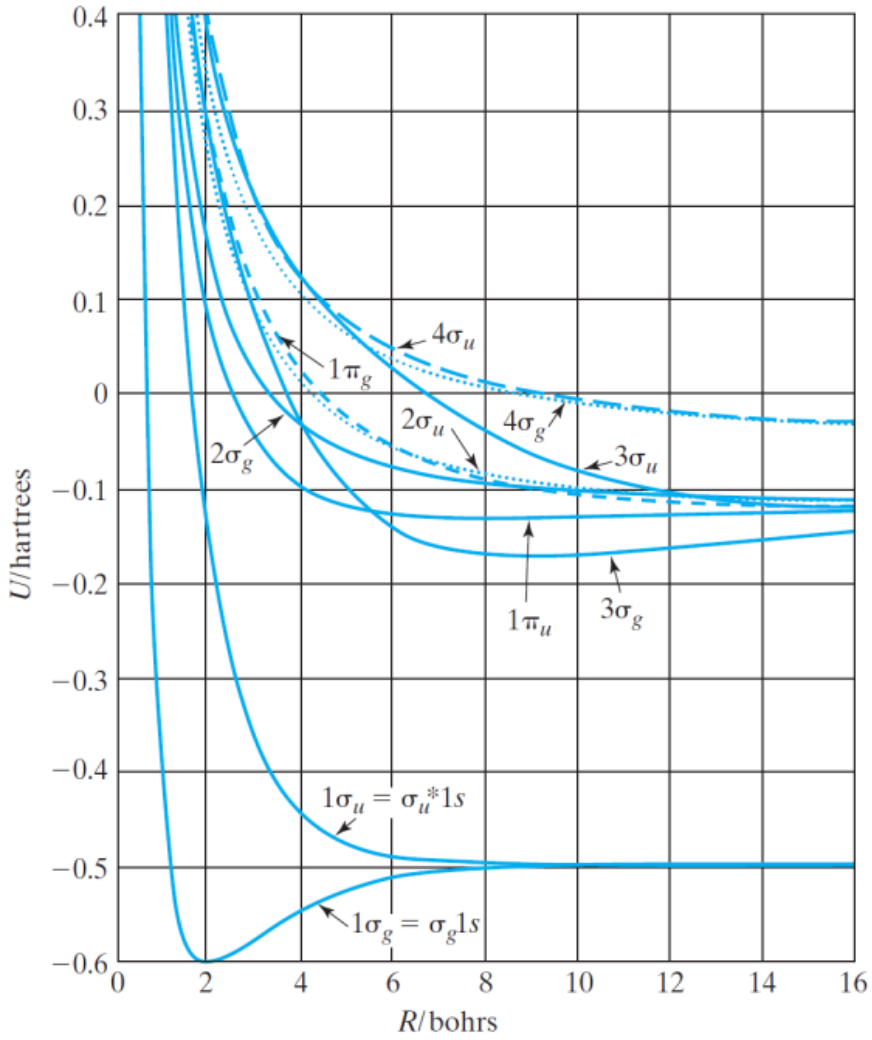


偶宇称(gerade)　奇宇称(ungerade)

镜面(反射reflection)对称性：关于垂直于键轴的中心平面的对称性，反对称的标*

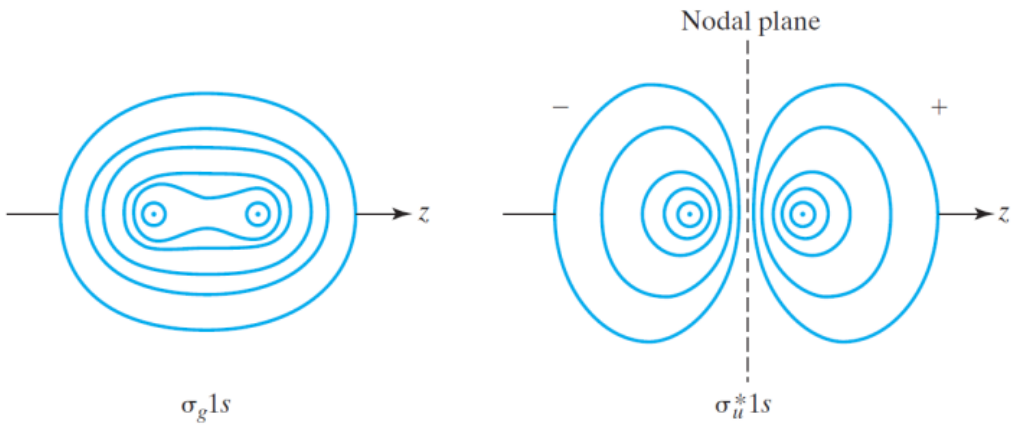
奇偶性与成键反键是两回事

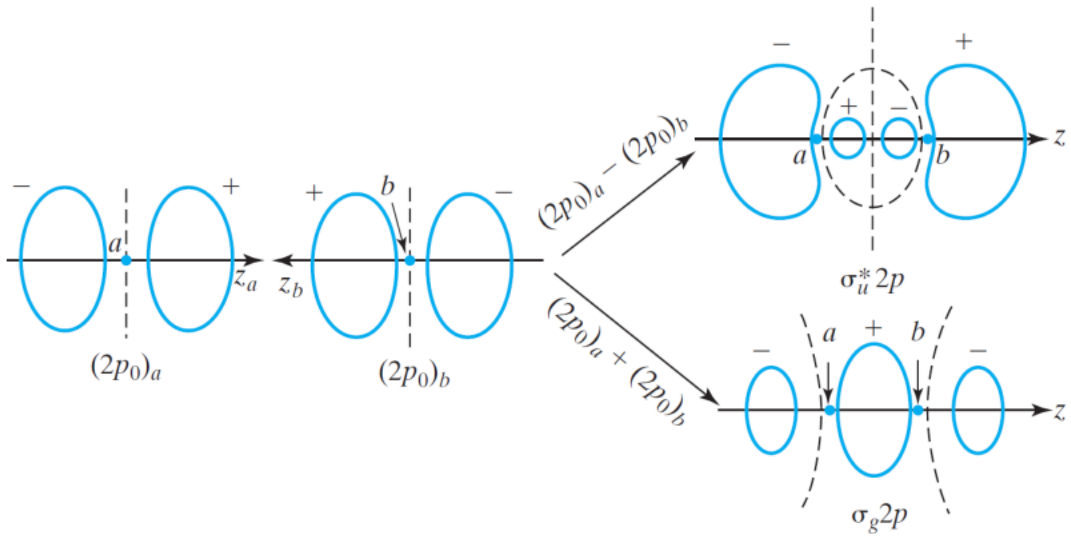
镜面反对称的是反键



H_2^+势能曲线（势能面）

σ轨道





LCAO：

$$\phi = c_1(1s_a \pm 1s_b)$$

$$\phi = c_2(2s_a \pm 2s_b)$$

$$(2p_0)_a \pm (2p_0)_b = (2p_z)_a \pm (2p_z)_b$$

参与组合的原子轨道
角动量z分量都为0

解析形式： $\psi = N(\xi)M(\eta)\frac{1}{\sqrt{2\pi}}e^{im\phi}$

$m = 0$ 轨道角动量z分量为0

关于节面：基态无节面，其他轨道的节面为椭球面或双曲面（可退化为平面）

π轨道

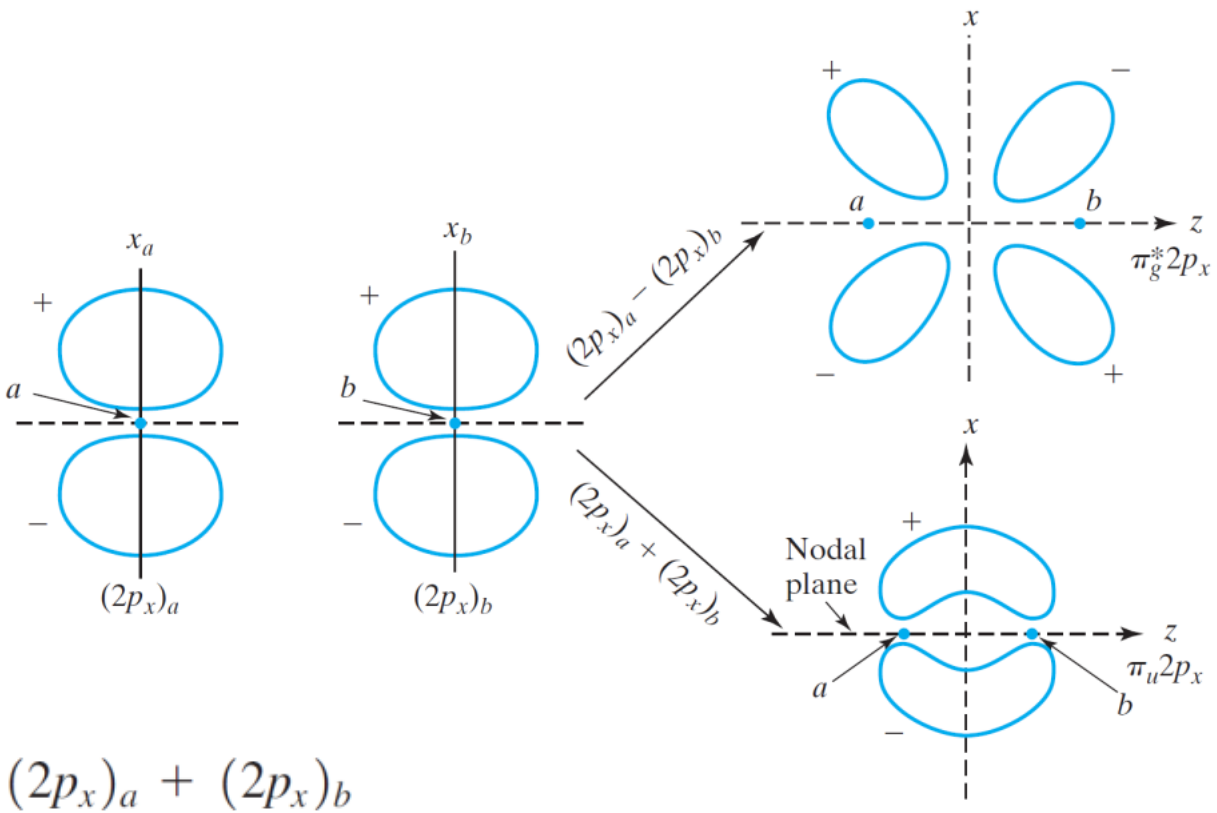
LCAO：

$$(2p_{+1})_a + (2p_{+1})_b$$
$$(2p_{+1})_a - (2p_{+1})_b$$
$$(2p_{-1})_a + (2p_{-1})_b$$
$$(2p_{-1})_a - (2p_{-1})_b$$

参与组合的原子轨道角动量z分量同为1或同为-1

进一步组合为实轨道：

$$(2p_x)_a + (2p_x)_b$$
$$(2p_x)_a - (2p_x)_b$$
$$(2p_y)_a + (2p_y)_b$$
$$(2p_y)_a - (2p_y)_b$$

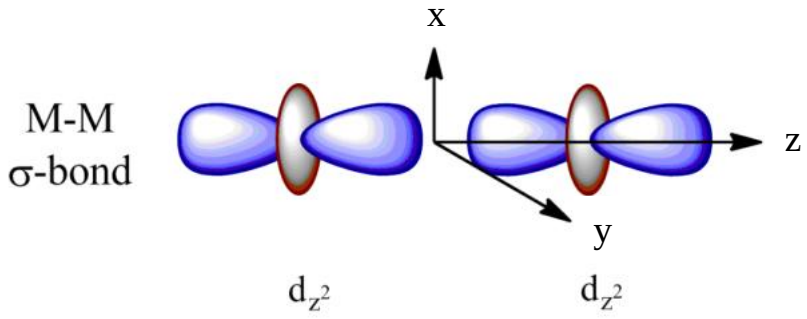


π轨道有沿着键轴的节面，如同p轨道

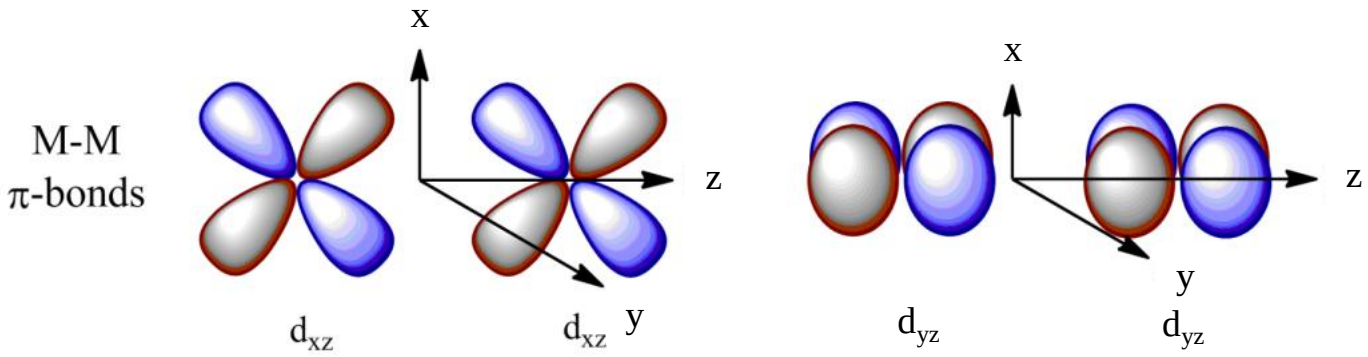
解析形式：$\psi = N(\xi)M(\eta)\frac{1}{\sqrt{2\pi}}e^{im\phi}$

$|m| = 1$ 轨道角动量z分量量子数绝对值为1

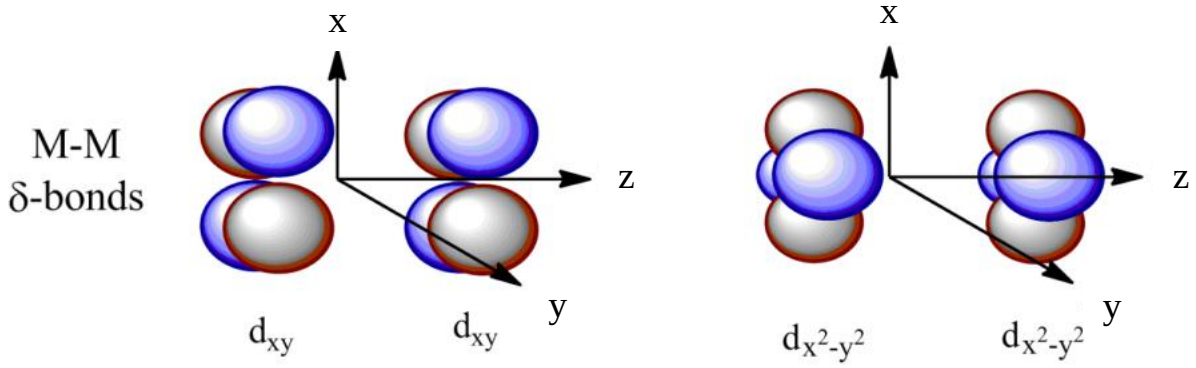
d轨道组合形成分子轨道



键轴方向角动量为0



键轴方向角动量量子数绝对值$|m|$为1

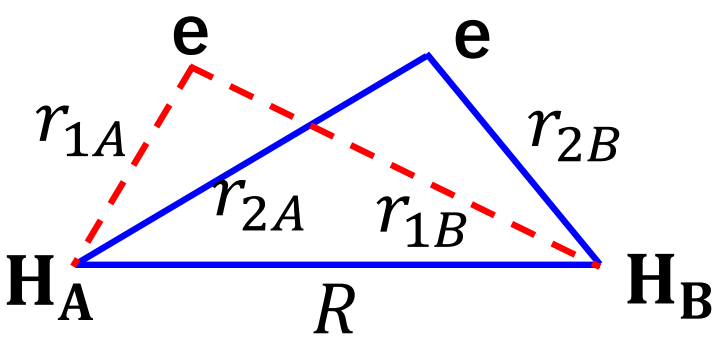


键轴方向角动量量子数绝对值$|m|$为2

修改自rachelleogyaz.blogspot.com

（三） 分子轨道理论

H_2分子薛定谔方程



$$\left(-\frac{1}{2}\nabla_1^2-\frac{1}{2}\nabla_2^2\right)\Psi+\left[-\frac{1}{|\boldsymbol{r}_1-\boldsymbol{R}_A|}-\frac{1}{|\boldsymbol{r}_1-\boldsymbol{R}_B|}-\frac{1}{|\boldsymbol{r}_2-\boldsymbol{R}_A|}-\frac{1}{|\boldsymbol{r}_2-\boldsymbol{R}_B|}+\frac{1}{|\boldsymbol{r}_1-\boldsymbol{r}_2|}\right]\Psi+\frac{1}{R}\Psi=E\Psi$$

$$\Psi=\Psi(\boldsymbol{r}_1,s_1;\boldsymbol{r}_2,s_2)$$

$$=\Psi(x_1,y_1,z_1,s_1;x_2,y_2,z_2,s_2)$$

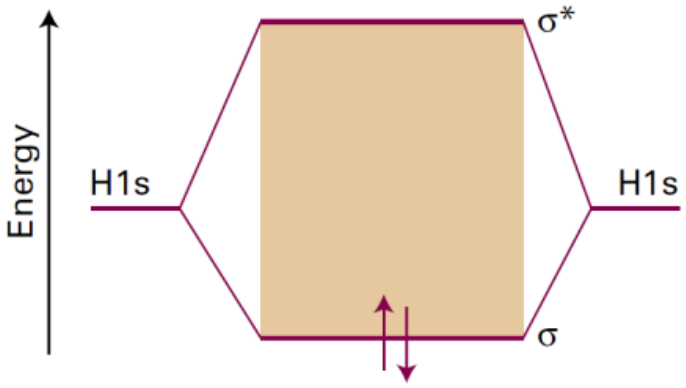
s_i是自旋维度， $s_i=\uparrow$ or $\downarrow$

- 多电子问题，用**单电子图像近似**
- N个电子填在N个分子轨道中
- **分子轨道理论**

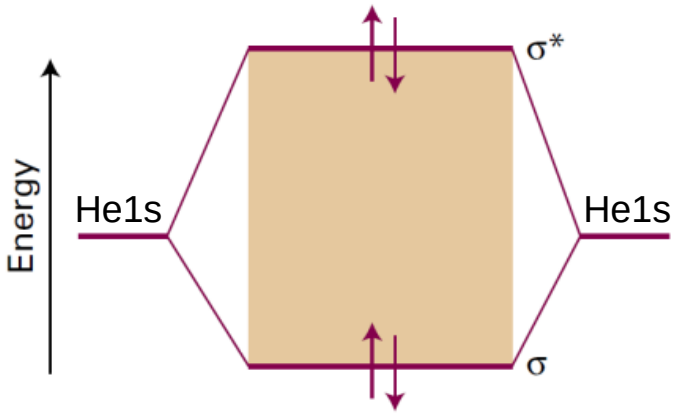
- 电子的交换对称性

$$\Psi(\boldsymbol{r}_1,s_1;\boldsymbol{r}_2,s_2)=-\Psi(\boldsymbol{r}_2,s_2;\boldsymbol{r}_1,s_1)$$

分子轨道能级图

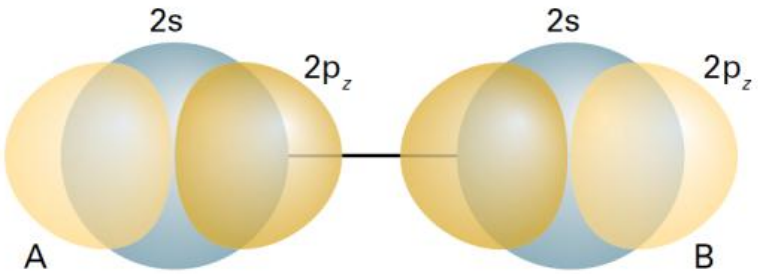


H_2

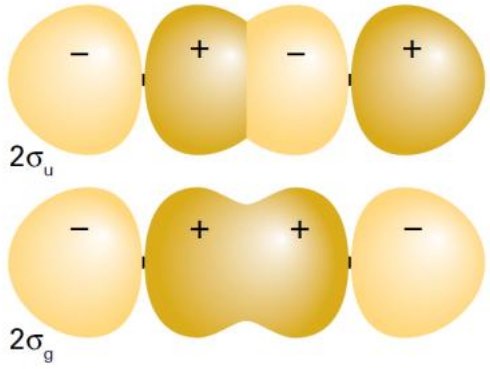


He_2

原子轨道组合时，遵从对称性匹配原则



2s与$2p_z$轨道角动量分量为0

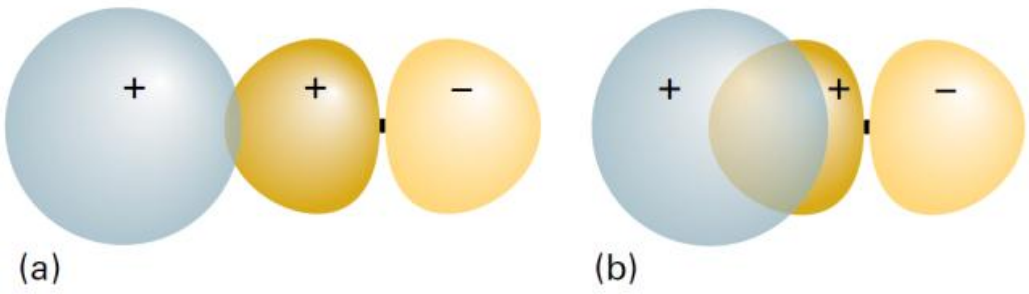


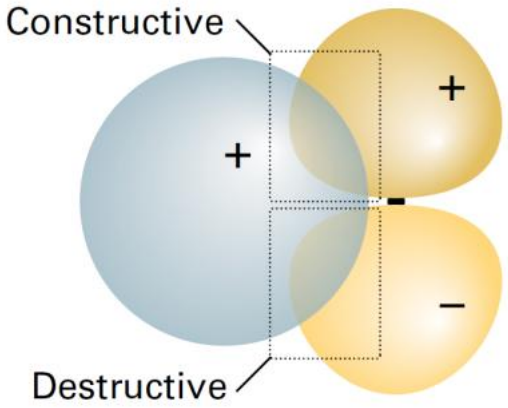
核间发生干涉，相长干涉，能量降低；相消干涉，能量升高

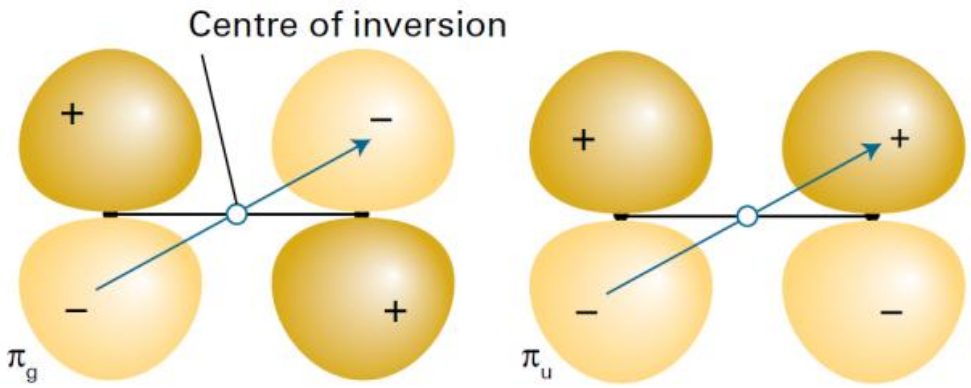
原子轨道交叠积分

$$S=\int \psi_A^{*}\psi_B \, d\tau$$

成键强度与交叠积分正相关



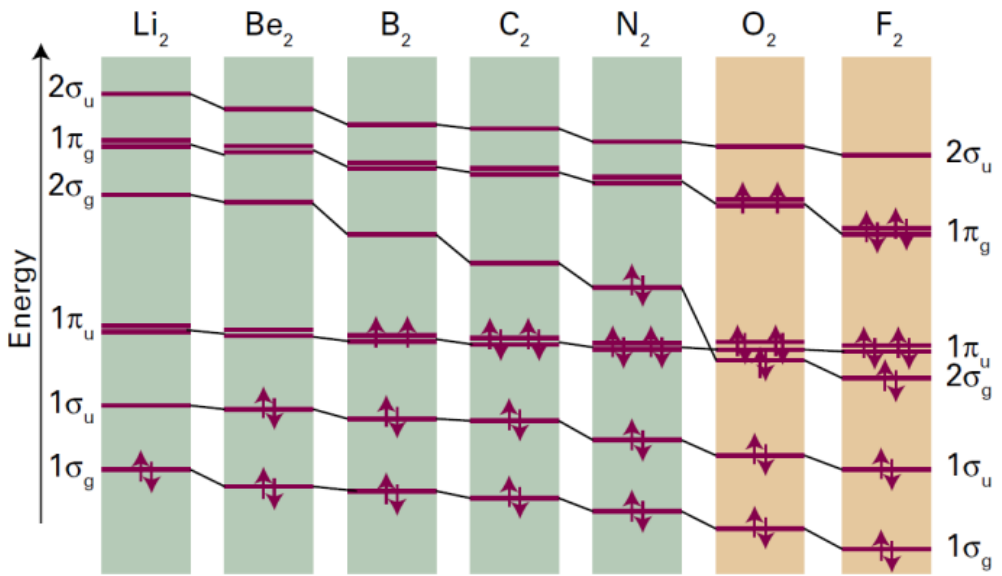




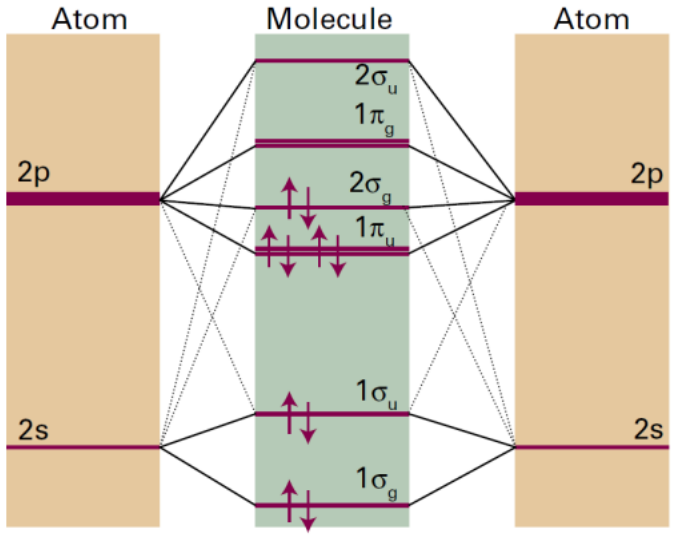
从轨道交叠角度也能解释s与p轨道无法形成π键

但从交叠解释σ键比π键弱比较牵强，见书

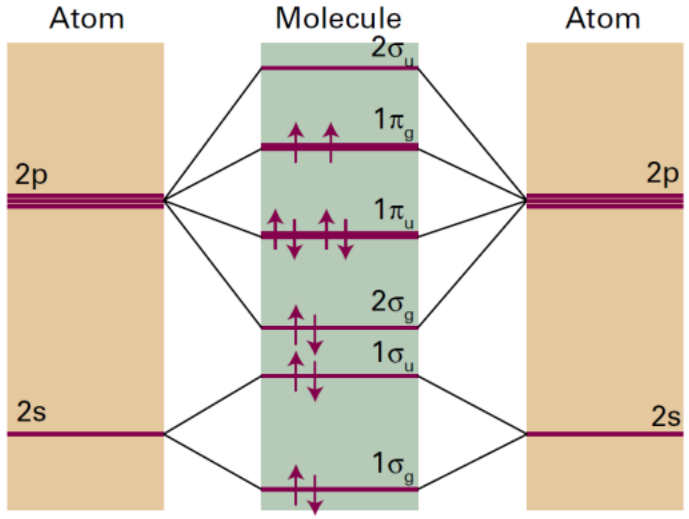
第二周期分子轨道能级图



- 分子轨道本身基于单电子假设，只能作为参考，不完全真实
- $1\pi_u$与$2\sigma_g$轨道能量顺序：没有什么道理，不同计算方法得到的结论也不一样



Li_2到N_2， $1\pi_u$低于$2\sigma_g$



O_2与F_2， $1\pi_u$高于$2\sigma_g$

键级的概念 $b=\frac{1}{2}(N-N^*)$

键级与键长、键能的关系

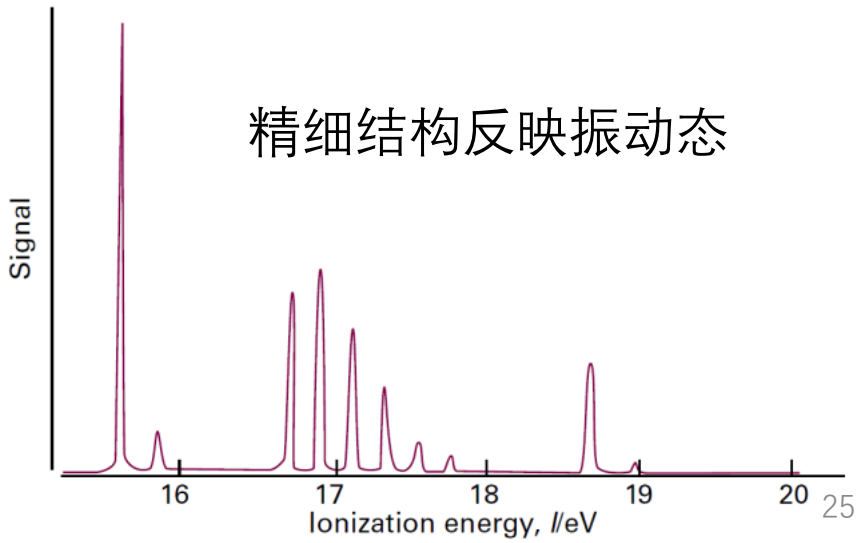
Bond	Order	R_e/pm
HH	1	74.14
NN	3	109.76
HCl	1	127.45
CH	1	*114*
CC	1	*154*
	2	*134*
	3	*120*

Bond	Order	$N_A hc\tilde{D}_0$/(kJ mol^{-1})
HH	1	432.1
NN	3	941.7
HCl	1	427.7
CH	1	*435*
CC	1	*368*
	2	*720*
	3	*962*

光电子能谱(photoelectron spectroscopy)

测量电离能，近似推出轨道能

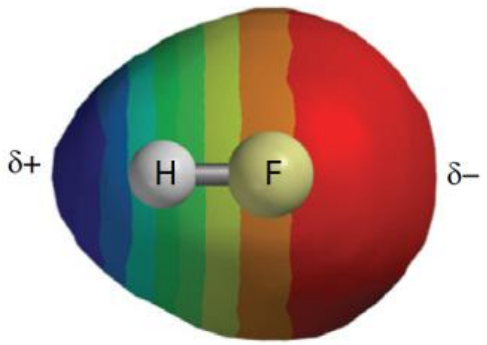
UPS：紫外(电离能一般不在可见光区)
XPS：X射线



异核双原子分子

极性键

电负性χ（electronegativity）：
反映原子吸引电子的能力



$$|\chi_A - \chi_B| = \{hc\tilde{D}_0(AB) - \tfrac{1}{2}[hc\tilde{D}_0(AA) + hc\tilde{D}_0(BB)]\}^{1/2}/\text{eV}$$

Pauling electronegativity [definition]

$$\chi = \tfrac{1}{2}(I + E_{ea})/\text{eV}$$

Mulliken electronegativity [definition]

Element	χ_P
H	2.2
C	2.6
N	3.0
O	3.4
F	4.0
Cl	3.2
Cs	0.79

两种电负性正相关
（关系不重要）

变分法（variation principle）

求$\hat{H}$的基态能量E_0

$$E = \frac{\langle\psi_0|\hat{H}|\psi_0\rangle}{\langle\psi_0|\psi_0\rangle} = \langle\psi_0|\hat{H}|\psi_0\rangle$$

有限基组下能量期望值表达式对c_n求极值即可得到变分解

任意试探波函数

用$\hat{H}$本征态表示

$$\psi = \sum_n c_n \psi_n$$

注意：本征态之间正交归一，ψ归一要求

$$\sum_n |c_n|^2 = 1$$

变分原理：

$$E \leqslant \langle\psi|\hat{H}|\psi\rangle$$

证明：

$$\langle\psi|\hat{H}|\psi\rangle = \sum_{jk} c_j^* c_k \langle\psi_j|\hat{H}|\psi_k\rangle = \sum_{jk} c_j^* c_k \langle\psi_j|E_k|\psi_k\rangle$$

$$= \sum_{jk} c_j^* c_k E_k \delta_{jk} = \sum_k |c_k|^2 E_k \geqslant \sum_k |c_k|^2 E_0 = E_0$$

用变分法求异核双原子分子轨道

LCAO: $$\psi = \sum_n c_n \psi_n$$

这里的原子轨道不正交，从而ψ的归一性条件复杂

改从以下能量期望值出发 $$E = \frac{\int \psi \hat{H} \psi \, d\tau}{\int \psi^2 \, d\tau}$$

假定每个原子贡献一个轨道

$$\int \psi^2 \, d\tau = \int (c_A \psi_A + c_B \psi_B)^2 \, d\tau$$

$$= c_A^2 \overbrace{\int \psi_A^2 \, d\tau}^{1} + c_B^2 \overbrace{\int \psi_B^2 \, d\tau}^{1} + 2 c_A c_B \overbrace{\int \psi_A \psi_B \, d\tau}^{S}$$

$$= c_A^2 + c_B^2 + 2 c_A c_B S$$

$$\int \psi \hat{H} \psi \, d\tau = \int (c_A \psi_A + c_B \psi_B) \hat{H} (c_A \psi_A + c_B \psi_B) \, d\tau$$

$$= c_A^2 \overbrace{\int \psi_A \hat{H} \psi_A \, d\tau}^{\alpha_A} + c_B^2 \overbrace{\int \psi_B \hat{H} \psi_B \, d\tau}^{\alpha_B} + c_A \; c_B \overbrace{\int \psi_A \hat{H} \psi_B \, d\tau}^{\beta}$$

$$+ c_A c_B \overbrace{\int \psi_B \hat{H} \psi_A \, d\tau}^{\beta}$$

$$\int \psi \hat{H} \psi \, d\tau = c_A^2 \alpha_A + c_B^2 \alpha_B + 2 c_A c_B \beta$$

$$E=\frac{c_A^2\alpha_A+c_B^2\alpha_B+2c_Ac_B\beta}{c_A^2+c_B^2+2c_Ac_BS}$$

对能量表达式变分求极值

$$\frac{\partial E}{\partial c_A}=0 \qquad \frac{\partial E}{\partial c_B}=0$$

$$\frac{\partial E}{\partial c_A}=\frac{2\{(\alpha_A-E)c_A+(\beta-SE)c_B\}}{c_A^2+c_B^2+2c_Ac_BS}$$

$$\frac{\partial E}{\partial c_B}=\frac{2\{(\alpha_B-E)c_B+(\beta-SE)c_A\}}{c_A^2+c_B^2+2c_Ac_BS}$$

最终推得久期方程（secular equations）

$$(\alpha_A-E)c_A+(\beta-SE)c_B=0$$
$$(\alpha_B-E)c_B+(\beta-SE)c_A=0$$

$$\alpha_A=\int\psi_A\hat{H}\psi_A\mathrm{d}\tau \qquad \alpha_B=\int\psi_B\hat{H}\psi_B\mathrm{d}\tau$$

Coulomb integrals

$$\beta=\int\psi_A\hat{H}\psi_B\mathrm{d}\tau=\int\psi_B\hat{H}\psi_A\mathrm{d}\tau$$

Resonance integral

$$S=\int\psi_A\psi_B\mathrm{d}\tau$$

Overlap integral

久期方程等价于久期行列式方程

$$\begin{vmatrix}\alpha_A-E & \beta-SE\\ \beta-SE & \alpha_B-E\end{vmatrix}=(\alpha_A-E)(\alpha_B-E)-(\beta-SE)^2=0$$

同核时($\alpha_A=\alpha_B$)，解可化简为

$$E_\pm=\frac{(\alpha\pm\beta)(1\mp S)}{(1+S)(1-S)}$$

原子轨道能量匹配时，成键/反键效应最强，分析见书

多原子分子

以乙烯p_z轨道形成π键为例

久期方程

$$\begin{vmatrix} \alpha - E & \beta - ES \\ \beta - ES & \alpha - E \end{vmatrix} = 0$$

$$\begin{vmatrix} \alpha - E & \beta - ES \\ \beta - ES & \alpha - E \end{vmatrix} \longrightarrow \begin{vmatrix} \alpha - E & \beta \\ \beta & \alpha - E \end{vmatrix}$$

休克尔近似

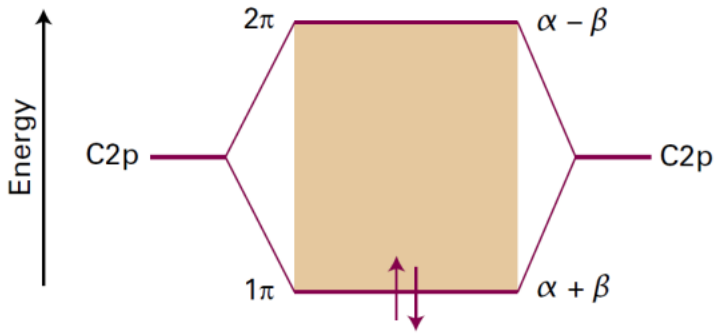
- **所有的 $S = 0$**
- **β只考虑最近邻项，其他设为零**

定义矩阵改写久期方程

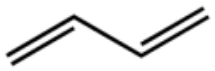
$$H = \begin{pmatrix} \alpha & \beta \\ \beta & \alpha \end{pmatrix} \qquad S = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$Hc = cE$$

$$\Longrightarrow \quad E = \alpha \pm \beta$$

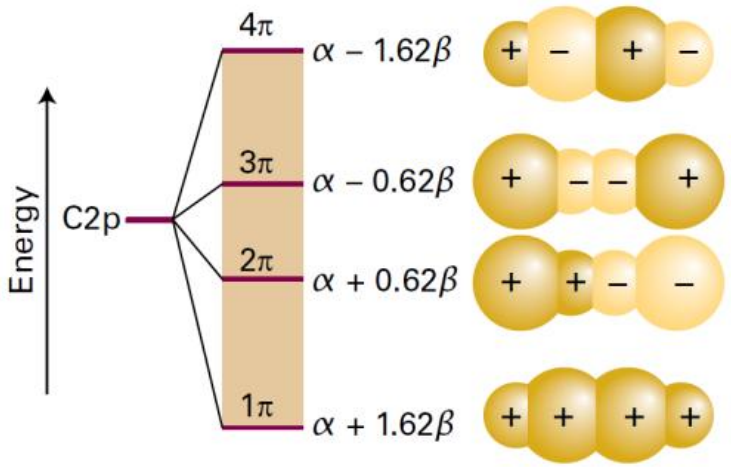


丁二烯的π轨道：离域分子轨道



1 Butadiene

$$H=\begin{pmatrix} \overbrace{H_{AA}}^{\alpha} & \overbrace{H_{AB}}^{\beta} & \overbrace{H_{AC}}^{0} & \overbrace{H_{AD}}^{0} \\ H_{BA} & H_{BB} & H_{BC} & H_{BD} \\ H_{CA} & H_{CB} & H_{CC} & H_{CD} \\ H_{DA} & H_{DB} & H_{DC} & H_{DD} \end{pmatrix} \xrightarrow{\text{Hückel approximation}} \begin{pmatrix} \alpha & \beta & 0 & 0 \\ \beta & \alpha & \beta & 0 \\ 0 & \beta & \alpha & \beta \\ 0 & 0 & \beta & \alpha \end{pmatrix}$$



对角化后得

$$E=\begin{pmatrix} \alpha+1.62\beta & 0 & 0 & 0 \\ 0 & \alpha+0.62\beta & 0 & 0 \\ 0 & 0 & \alpha-0.62\beta & 0 \\ 0 & 0 & 0 & \alpha-1.62\beta \end{pmatrix} \qquad c=\begin{pmatrix} 0.372 & 0.602 & 0.602 & 0.372 \\ 0.602 & 0.372 & -0.372 & -0.602 \\ 0.602 & -0.372 & -0.372 & 0.602 \\ 0.372 & -0.602 & 0.602 & -0.372 \end{pmatrix}$$

$$E_1=\alpha+1.62\beta \quad \psi_1=0.372\psi_A+0.602\psi_B+0.602\psi_C+0.372\psi_D$$
$$E_2=\alpha+0.62\beta \quad \psi_2=0.602\psi_A+0.372\psi_B-0.372\psi_C-0.602\psi_D$$
$$E_3=\alpha-0.62\beta \quad \psi_3=0.602\psi_A-0.372\psi_B-0.372\psi_C+0.602\psi_D$$
$$E_4=\alpha-1.62\beta \quad \psi_4=0.372\psi_A-0.602\psi_B+0.602\psi_C-0.372\psi_D$$

π键键能(休克尔近似模拟)

乙烯 $E_\pi = 2(\alpha+\beta) = 2\alpha + 2\beta$

丁二烯 $E_\pi = 2(\alpha+1.62\beta) + 2(\alpha+0.62\beta) = 4\alpha + 4.48\beta$

离域化能：0.48β

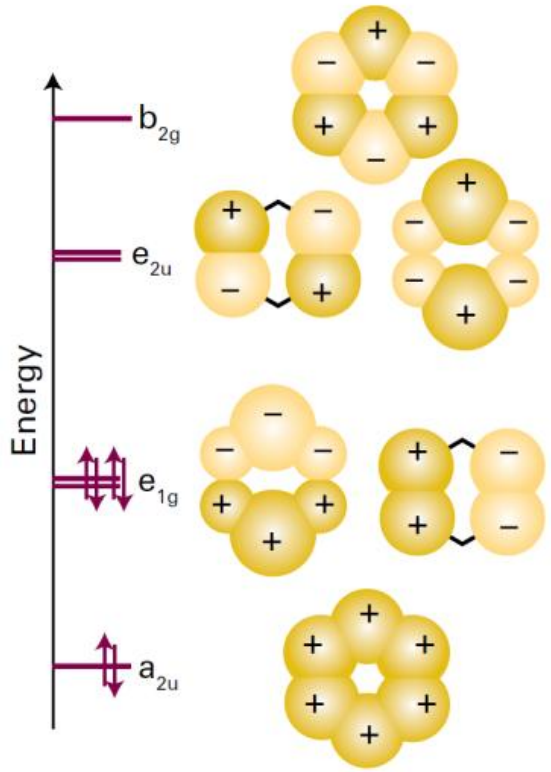
π键形成能： 4.48β $E_{bf} = E_\pi - N_C\alpha$

苯分子：

$$H = \begin{pmatrix} \alpha & \beta & 0 & 0 & 0 & \beta \\ \beta & \alpha & \beta & 0 & 0 & 0 \\ 0 & \beta & \alpha & \beta & 0 & 0 \\ 0 & 0 & \beta & \alpha & \beta & 0 \\ 0 & 0 & 0 & \beta & \alpha & \beta \\ \beta & 0 & 0 & 0 & \beta & \alpha \end{pmatrix}$$

$$E = \alpha \pm 2\beta, \alpha \pm \beta, \alpha \pm \beta$$

复习芳香性的概念



$$E_\pi = 2(\alpha+2\beta) + 4(\alpha+\beta) = 6\alpha + 8\beta$$

“原子轨道理论”总结

- 单电子原子（类氢原子），核库伦势

  严格定义 s, p, d…

- 多电子原子，引入有效势$v_{\mathrm{eff}}(r)$
- 径向方程改变，**角向方程不变**，仍可定义 s, p, d…

  1. 中心力场法 $v_{\mathrm{eff}}(r) = -\frac{Z-\sigma}{r}$
  2. 自洽场法 $v_{\mathrm{eff}}(r)$依赖于求解出的轨道

分子轨道理论总结

- 单电子分子（类H_2^+），核库伦势

  严格定义 σ, π, δ…

- 多电子分子，引入有效势$v_{\mathrm{eff}}(\boldsymbol{r})$
- ξ与η的方程改变，**ϕ的方程不变**，仍可定义 σ, π, δ…

  1. 自洽场法 $v_{\mathrm{eff}}(\boldsymbol{r})$依赖于求解出的轨道
  2. 有效哈密顿量 如休克尔方法